



MEETING WITH PROF. LUIS · 23 JUL 2026

Points to discuss with Luis

Follow-up to the 17 July meeting with Tulio: commitments, progress, and open decisions.

PUBLICATION 1

DSO-Led Fair Dynamic Operating Envelopes for Aggregator- and Prosumer-Level Flexibility Coordination

PRESENTER

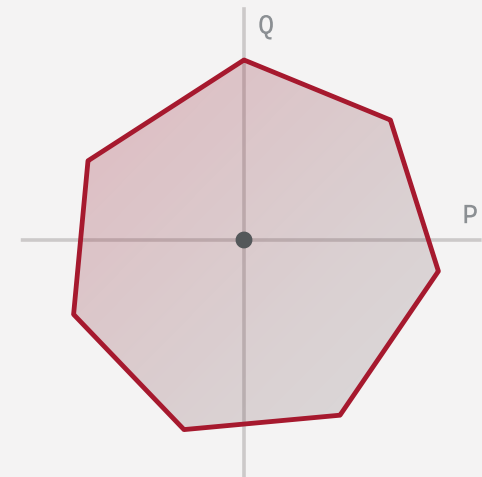
Ignacio Pérez Ortiz

TARGET

IEEE Trans. Smart Grid

NEXT CHECKPOINT

7 Aug 2026



Feasible Operating Region · P-Q plane

 WHAT I'M BRINGING TODAY

Five fronts for the meeting

- | | | |
|----------|---|---|
| A | Introduction — comments and commitments | Aligned on the storyline; tasks due 7 Aug |
| B | Methodology — commitments | Deep review and improvement proposals |
| C | Model progress | ≈ 2 weeks ahead of schedule |
| D | Key questions for Luis | Scope, aggregation, visualization |
| E | Case-study proposal | 1 / 2 / 9 transformers |

Aligned on the **storyline**; paragraph-by-paragraph commitments

PARAGRAPH	COMMITMENT · ALL DUE 7 AUG	OWNER
P1 + P2	Merge them; state it more directly and concisely.	Ignacio
P3	+2 references (from 2 to 4): coincident DER injections can cause network problems → aggregator offers must be network-feasible (the DSO's role).	Ignacio
P4 · FORs	Add more references (Tulio's area of expertise).	Tulio
Transition P4→P5	Smooth FORs → DOEs: a time-varying P-Q region maps naturally onto another time-varying P-Q region (the DOEs).	Ignacio
Transition fairness → multi-aggregator	Smooth the jump between the two paragraphs.	Ignacio
P7 · gap	Support the research gap with a table of ~5–6 key publications: each tackles part of the problem, none integrates the full framework.	Both
P9 · contributions	Rewrite the contributions as bullet points, not a running paragraph.	Ignacio

→ Every publication in the comparison table must also be cited/discussed in the introduction.

B METHODOLOGY

Validated P1–P2 structure; my task is to review it in depth

P1 · SELF-CONSUMPTION DISPATCH

Optimize the operating trajectory, execute the current-slot battery action, and propagate the measured/realized storage state.

P2 · REACHABLE NEXT-INTERVAL FOR

Compute the next-interval P-Q region while fixing the committed action and the energy state inherited from P1.

 My commitment (due 7 Aug)

Read the methodology in depth, insert comments, and propose improvements / ways to make it more concise — symmetric to what Tulio does with my introduction.

 Agreed acceleration

Start the multi-aggregator modeling earlier than planned, using Tulio's model and documentation.

C MODEL PROGRESS

≈ 2 weeks

ahead of the original schedule —
Tulio already committed/pushed, so
I could start early.

- ✓ **Repo recreated & synced**
Cloned + pulled Tulio's model and documentation; working in the shared workspace.
- ✓ **Setup validation — TR9 run**
Reproducing Tulio's baseline (PV, demand, export/import, SCR/SSR) to confirm my environment yields identical results.
- ✓ **Portable script routing**
Post-processing Python scripts moved to relative/portable paths (routing refactor) — committed and pushed.
- ✓ **Collaboration & git workflow rules**
CONTRIBUTING guide submitted, under Tulio's review for merge.
- ✓ **README with documentation rules**
Where each document lives (repo / local / OneDrive) to keep code separate from personal notes.
- ✓ **Compute-time improvement ideas**
Analysis of what AIMMS does outside the solver; a proposal to review with Tulio.
- ⊖ **Next: cross-validation with OpenDSS**
Check the model's results against an independent power flow.

D KEY QUESTIONS

What we need to resolve with Luis

Everything hinges on **compute time** and **paper space** (~10–11 pp limit): compact case studies, but analysis that delivers the value we want to show.

1 How many transformers in the case study?

The OPF runs per transformer (all together is infeasible) and the FOR is aggregated via Minkowski. Is 9 worth it, or is 2 or 1 enough? The value is the framework and analysis, not a huge realistic network → compact network, with simulation efficiency as future work.

2 How to disaggregate when aggregators span transformers?

If we compute the rolling FORs per transformer (in parallel) and sum them into a large-scale FOR, how do we disaggregate the PQ-DOEs when an aggregator spans different transformers? Extra reason to keep few transformers.

3 Which formulation level goes in the main paper?

Given the page limit: which OPF equations and figures stay in the main body vs the appendix. Confirm the journal's appendix options.

4 How to assign prosumers to the two aggregators?

Define an electrically meaningful criterion (near vs remote, different voltage sensitivities) rather than a purely random one. → **detailed scenarios proposed in E.**

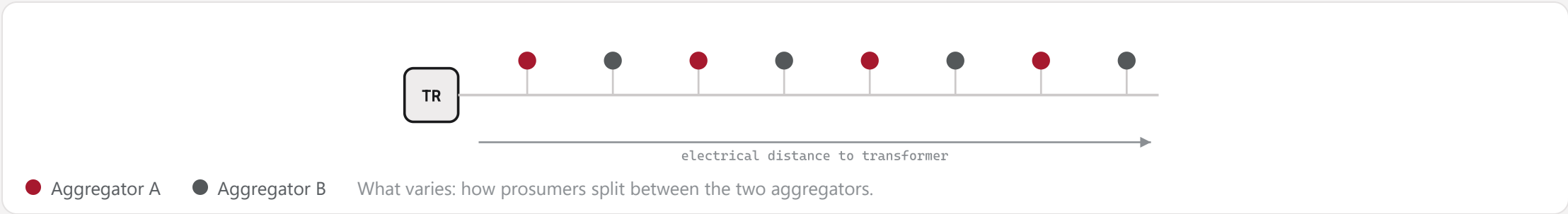
5 How to visualize the time-varying results?

How to present the FORs and PQ-DOEs and how they change under different fairness criteria / prosumer distributions. Proposed approach: the time-varying operating-region representation of Lankeshwara et al. (2024), already implemented by Tulio (with example figures).

E CASE-STUDY PROPOSAL

Initial scenarios — one proposal per transformer-count decision

One proposal per possible decision on the number of transformers (Q1: **1 / 2 / 9**, or another). For **each fairness criterion** at both levels, these are the prosumer→aggregator assignment scenarios — fewest scenarios, richest analysis.



CASE	ASSIGNMENT SCENARIOS · SPATIAL SPLIT BETWEEN AGGREGATORS
1 transformer	1 Uneven split by electrical distance to the transformer (e.g. one aggregator closer, the other farther). 2 Random, roughly balanced split.
2 transformers	1 One aggregator per transformer — vary the spatial spread of prosumers within each. 2 Mixed across both transformers — vary each aggregator's share per transformer and the prosumers' distance-to-transformer profile.
9 transformers	→ Same logic as the 2-transformer case (mixed), scaled up: shares per transformer and distance profiles.

Goal: an exact, compact set of scenarios that reveals how the **FORs** and **PQ-DOE assignment** change with **prosumer location** and fairness — minimum scenarios, maximum interpretability.

 NEXT STEPS

On the way to the 7 August checkpoint

MY DELIVERABLES BY 7 AUG

- Intro: merge P1+P2, +2 refs in P3, transitions (FORs→DOEs, fairness→multi-aggregator), contributions as bullets.
- Methodology: comments and improvement proposals.
- Start the multi-aggregator modeling.

TO CLOSE WITH LUIS TODAY

- Number of transformers for the case study.
- PQ-DOE disaggregation across transformers.
- Formulation level: main body vs appendix.
- Prosumer-assignment criterion.
- Time-varying visualization approach.

→ Guiding principle: prioritize **depth and interpretability** over the number of cases. · Tulio on vacation 27–31 Jul.